MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2026

FIFTH SEMESTER

ELECTIVE – MG5DSEBTG302 BIOPROCESS TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A – One word answer - Answer any 10 questions. Each question carries 2 marks

1. Name the scientist who discovered fermentation. [K] /CO1

2. What is the primary screening technique used to isolate antibiotic producing organisms? [K] / CO2

3. Give an example for a solid state fermentation. [K] / CO1

4. Write any two examples for secondary metabolites. [K] / CO1

5. Name the type of culture where critical components are added at regular intervals. [K] /CO1

6. Define auxotrophs. [U] / CO2

7 What is hybridisation? [U] / CO2

8. Define inoculum media. [U] / CO3

9. Cite an example for a precursor used in antibiotic production media [U] / CO3

10. Name the common sterilisation technique used for the sterilisation of fermentation media. [U] / CO3

11. Define baffles. [U] / CO3

12. Name the salt used for the purification of proteins. [U] / CO4

(10X2=20)

Part B- Short essay- Answer any 4 questions. Each question carries 5 marks

13. Describe auxanography. [A] / CO2

14. What are the applications of Bioprocess Technology? [A] / CO1

15. Write notes on fermentation . [U] / CO1

16. Explain the parts of a bioreactor. [U] / CO3

17. What are the different solid- liquid separation techniques used in downstream processing? [K] / CO4

18. Write in detail on the various cell immobilisation techniques. [A] / CO4

(4X5=20)

Part C- Essay- Answer any 2 question. Each question carries 15 marks

19. Write a detailed account on the various mutation techniques used in strain improvement. [A] / CO2

20. Write an essay on the different steps in downstream processing. [A] /CO3

21. Write notes on the fermentative production of penicillin. [K] / CO4

22. Write in detail different types of bioreactors. [A]/CO3

(2X15=30)

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